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Project Description

SafeTrade provides an intuitive application that securely connects lenders and borrowers, enables negotiable rental terms, ensures financial security for both parties, and offers a seamless user experience. SafeTrade aims to create a reliable and efficient marketplace for item lending within local communities.

Project Deliverables

An MVP will require the following functionalities:

1. Secure account creation allows users to register and sign in.
2. Profile management functionality for the storage and retrieval of user metadata, including names, contact details, and localised geographic information.
3. Asset listing capabilities enabling users to create and manage detailed item catalogues with support for unstructured media (photos), pricing structures, and availability schedules.
4. A spatial and categorical discovery engine that allows users to search for rental items by proximity and filter by parameters such as price range and rental duration.
5. A transactional messaging interface facilitating peer-to-peer communication, allowing borrowers to initiate rental requests and negotiate terms directly with lenders.
6. A reporting system for dispute resolution, allowing users to submit tickets supported by evidence for platform moderation.
7. A notification system that monitors the item catalogue and triggers alerts when assets become available within a price threshold set by the user.

Quality Attributes

Below are the main quality attributes the application will focus on. In addition to these, implement strong privacy and security measures to protect user data.

1. **Availability:** Reliability is crucial to SafeTrade, as users expect a dependable, consistently available platform for their lending and borrowing needs. The platform should be designed for high availability, ensuring that users can consistently access the service, view listings, and complete transactions without errors or downtime. The platform will be thoroughly tested and monitored for any potential issues or failures.
2. **Extensibility:** Extensibility is important for SafeTrade to accommodate future growth, new features, and integrations. As the platform evolves and user needs change, the ability to easily extend and modify the system architecture becomes crucial. An extensible architecture enables SafeTrade to adapt to market demands, introduce new functionality, and integrate with external systems and services.

3. Scalability: The platform's scalability will be evaluated based on its ability to maintain performance and data integrity as the user base, asset catalogue, and concurrent traffic grow. Evidence must show its capacity to handle high volumes of concurrent photo uploads for new listings and search requests.